■# PAPER<i>350 — El Gordo (ACT-CL J0102-4915): Most Massive z>0.5 Cluster — Super-Virial Merger F</i>U<i>Bi</i>i

Author: Daniel T. Murphy

Session: *96*

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 96 **Source:** gok*share*31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF F*U*Bi_i for El Gordo — highest-mass z>0.5 merger cluster **Author:** Daniel T. Murphy

<!-- UQFF constants: κ = 5.0e-4 day^{-1}, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

El Gordo (ACT-CL J0102-4915) is the most massive known galaxy cluster at z > 0.5, with M = 3×10^{1}■ M■ and a super-virial merger velocity Δv = 2500 km/s — more than double the cluster's virial velocity dispersion. The UQFF buoyancy-unified force yields F*U*Bi*i ≈ −1.40×10^{21}■ N, matching SPT-CL J2215 in the HIGHEST F*U*Bi*i tier. The super-virial velocity exceeds the standard ΛCDM prediction, and the UQFF provides an alternative mechanism: enhanced vacuum buoyancy accelerates the merger beyond the virial limit.

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2. Core Physics

2.1 UQFF Buoyancy-Unified Force

```
F_{U\_Bi\_i} \approx -1.40 \times 10^{218}\ \mathrm{N}
```

2.2 Super-Virial Merger Velocity

The virial velocity for El Gordo:

```
\sigma_v = \sqrt{\frac{GM_{\rm cl}}{R_{\rm cl}}} \approx 1100\ \mathrm{km/s}
```

The observed Δv = 2500 km/s exceeds this by factor ~2.27. The UQFF explanation:

```
\Delta v_{\rm UQFF} = \sigma_v + v_{\rm buoyancy}
```

```
v_{\rm buoyancy} = \sqrt{\frac{2 F_{U\_Bi\_i}}{M_{\rm cl}}} \cdot |t_{\rm merge}|
```

2.3 Cluster Mass

```
M_{\rm cl} = 3 \times 10^{15}\ M_\odot = 5.97 \times 10^{45}\ \mathrm{kg}
```

This is approximately 10× the SPT-CL J2215 mass in absolute terms, yet both yield similar F*U*Bi*i, indicating that F*U*Bi*i is not purely mass-dependent at cluster scales.

2.4 Redshift Context

```
z = 0.87 \quad \Rightarrow \quad x_2 \approx 5.6\ \mathrm{Gly}
```

At z = 0.87 the Universe was 53% of its current age. El Gordo could only have formed through non-standard mechanisms if ΛCDM is the only framework — UQFF vacuum buoyancy provides the required additional acceleration.

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3. Key Values

Quantity	Formula	Value
z	ACT/Planck	0.87
M_cl	Sunyaev-Zel'dovich	3×10^{1}■ M■
Δv	Spectroscopic merger	2500 km/s
σ_v (virial)	√(GM/R)	~1100 km/s
Δv/σ_v (super-virial ratio)	—	×2.27
F*U*Bi_i	UQFF full	-1.40×10^{21}■ N
x_2	Comoving	~5.6 Gly

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4. Physical Significance

El Gordo represents the "impossible cluster" problem in ΛCDM cosmology — its mass and merger velocity at z = 0.87 are exceedingly unlikely in standard cold dark matter simulations. The UQFF provides a natural explanation: F*U*Bi*i ≈ -1.40×10^{21}■ N is the additional vacuum buoyancy force that accelerates the merger, transforming a sub-virial encounter into a super-virial one. The same FUBi*i magnitude as SPT-CL J2215 (despite 4× higher mass) demonstrates UQFF saturation behavior at cluster mass scales.

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5. Deduplication Note

vs. PAPER_349 (SPT-CL J2215): Both yield F_U_Bi_i ≈ -1.40×10^{21}■ N; different physics drives the enhancement (SFR cool core vs. super-virial merger velocity).

Unique: Super-virial Δv = 2500 km/s is unique to El Gordo in the UQFF dataset.

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6. Classification

Physics Territory: FIRST UQFF super-virial merger with F*U*Bi_i ≈ -1.40×10^{21}■ N **Scale:** Cosmological (M ~ 10^{1}■ M■, z = 0.87) **CP Implementation:** `ElGordoACTCLJ0102MergerFUBiCalculator` (CondensedPhysics3.py, Session 96)
